

High-Bandwidth Serpentine Segmented Silicon Photonic Mach-Zehnder Modulator for 192 Gbaud Transmission

Qian Liu^{1,2}, Chao Cheng^{1,2}, Yangming Ren¹, Xinyu Li¹, Wenfu Zhang^{1,2}, and Binhao Wang^{1,2*}

¹State Key Laboratory of Ultrafast Optical Science and Technology, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China

²School of Future Technology, University of Chinese Academy of Sciences, Beijing 100049, China
*wangbinhao@opt.ac.cn

Abstract: We demonstrate a serpentine segmented silicon Mach-Zehnder modulator with a 1 dB bandwidth >67 GHz at 0 V bias and a modulation efficiency of 1.14 V·cm, supporting 192 Gbaud transmission.

1. Introduction

Driven by the explosive growth of artificial intelligence, high-performance computing, and cloud computing, global data traffic is expanding at an unprecedented pace, with data rates advancing toward 1.6 Tbps and beyond. Silicon photonics, offering low cost, high integration, and low power consumption, has emerged as a key technology to address these transmission bottlenecks [1]. As the core component enabling the conversion between electrical and optical signals in communication systems, the electro-optic (EO) modulator plays a crucial role [2]. With the continuous growth in communication capacity and bandwidth density, modulators are facing increasingly stringent demands on speed, power efficiency, and integration [3].

Silicon photonic modulators primarily rely on the plasma dispersion effect [4] within PN junctions. However, due to the relatively weak effect in silicon, a long modulation region is often required to achieve low-voltage operation. This constraint not only limits the EO bandwidth but also poses challenges for compact integration. To overcome these limitations, various modulator architectures have been proposed to enhance performance. For instance, a $124\text{ }\mu\text{m}$ -long silicon modulator leveraging the slow-light effect of Bragg gratings demonstrated a 3 dB bandwidth of 110 GHz and supported OOK transmission at 112 Gbaud [5]. In another approach, a segmented electrode silicon modulator achieved a 3 dB bandwidth exceeding 67 GHz, enabling 120 Gbaud 8ASK signaling [6]. An EO bandwidth above 110 GHz has also been reported using tunable time-frequency equalization technique [7]. Furthermore, impedance peaking in silicon traveling-wave modulators enables bandwidths up to 63.5 GHz [8], while distributed micro-capacitor designs can significantly extend the bandwidth without increasing the $V\pi$ compared with conventional structures [9]. Although these approaches have considerably advanced modulator performance, further enhancement in speed, efficiency, and integration remains a critical research focus.

To enable high-speed transmission within a compact footprint, we demonstrate a serpentine segmented-electrode Mach-Zehnder modulator (MZM) at O-band. The proposed device occupies only half the area of a conventional MZM with the same length, exhibiting a modulation efficiency of 1.14 V·cm and a 1 dB EO bandwidth exceeding 67 GHz at 0 V bias, supporting 192 Gbaud OOK transmission.

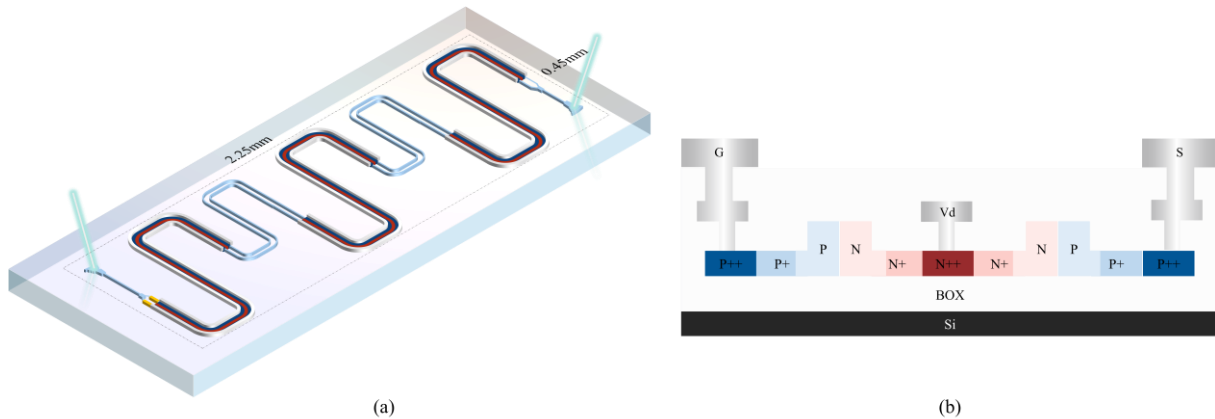


Fig. 1. (a) Schematic of the serpentine segmented MZM. (b) Cross-sectional view of the serpentine segmented MZM.

2. Device Design

Fig. 1(a) illustrates the schematic of the proposed serpentine segmented MZM structure. The device consists of three modulation segments with serpentine-arranged ground-signal (GS) traveling-wave electrodes and waveguides. Each phase shifter has a length of 950 μm and the overall modulator footprint is 2.25 mm $\times$ 0.45 mm. This configuration is designed for flip-chip integration with a distributed driver chip, which significantly enhances integration density and enables higher bandwidth compared with conventional wire-bonded photonic-electronic integration. The device is fabricated on a silicon-on-insulator (SOI) platform with a 220 nm-thick top silicon layer and a 2 μm -thick buried oxide (BOX). The cross-sectional structure of the modulator phase shifter is illustrated in Fig. 1(b). The phase shifters employ laterally structured PN junctions with three-level doping concentrations in 220 nm $\times$ 380 nm silicon rib waveguides with a 150 nm etch depth. To optimize device performance, the P and N regions are doped to $6 \times 10^{17} \text{ cm}^{-3}$ and $1 \times 10^{18} \text{ cm}^{-3}$, with a 100 nm junction offset introduced to balance optical loss and modulation efficiency. Intermediate p⁺ and n⁺ doping concentrations of $1 \times 10^{19} \text{ cm}^{-3}$ are used to simultaneously mitigate optical loss, PN-junction-limited bandwidth, and microwave attenuation, while heavily doped p⁺⁺ and n⁺⁺ regions ($1 \times 10^{20} \text{ cm}^{-3}$) form the ohmic contacts. The modulator adopts a series push-pull electrode configuration, with mismatched terminating resistors (40 Ω) integrated at the end of the 52 Ω traveling-wave electrodes to enhance bandwidth. A 100 μm optical path imbalance is introduced between the two modulation arms to facilitate DC characterization and bias tuning via wavelength adjustment. Additionally, thermal phase shifters are integrated on both arms to precisely control the bias point and maintain optimal linearity.

3. Experimental Results

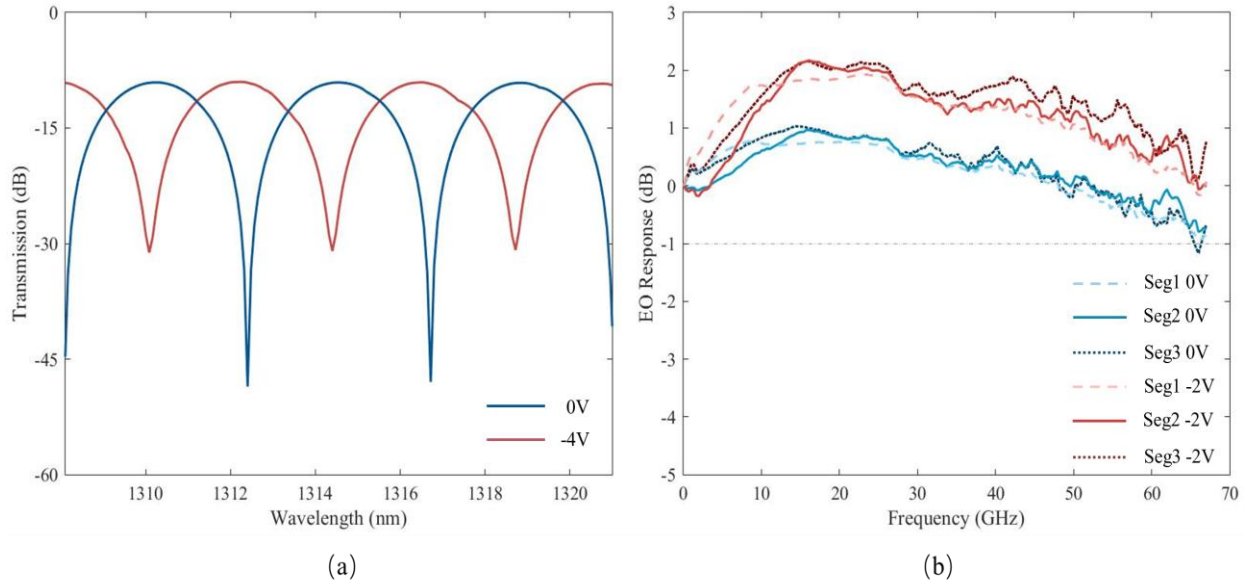


Fig. 2. (a) Measured transmission spectra of the modulator at 0 V and -4 V. (b) EO responses of the three segments at 0 V and -2 V.

Fig. 2(a) shows the static transmission spectra of the modulator when all segments are driven simultaneously, revealing a measured insertion loss of 9 dB. To extract the modulation efficiency, one junction was bypassed while monitoring the resulting phase shift, and procedure was repeated for the other junction. The average of the two measurements yields a $V\pi$ of 4 V under a -2 V bias, corresponding to a modulation efficiency of 1.14 V $\cdot$ cm. To evaluate the high-frequency performance, the small-signal EO response was measured using a 67 GHz photonic network analyzer (PNA) and a 100 GHz photodetector. The optical signal was coupled in and out of the modulator via grating couplers, while RF signal was applied to one segment of the modulator through a 67 GHz GS RF probe. A bias-tee was used to set the DC offset of the modulation signal. Fig. 2(b) presents the measured EO responses of three modulation segments at bias voltages of 0 V and -2 V. Because of the mismatched terminating resistors, a low-frequency peaking effect was observed. Under a 0 V bias, all segments exhibit a 1 dB EO bandwidth exceeding 67 GHz. Moreover, the bandwidth of all segments shows almost no degradation when a -2 V bias is applied, indicating a 3 dB EO bandwidth exceeding 100 GHz.

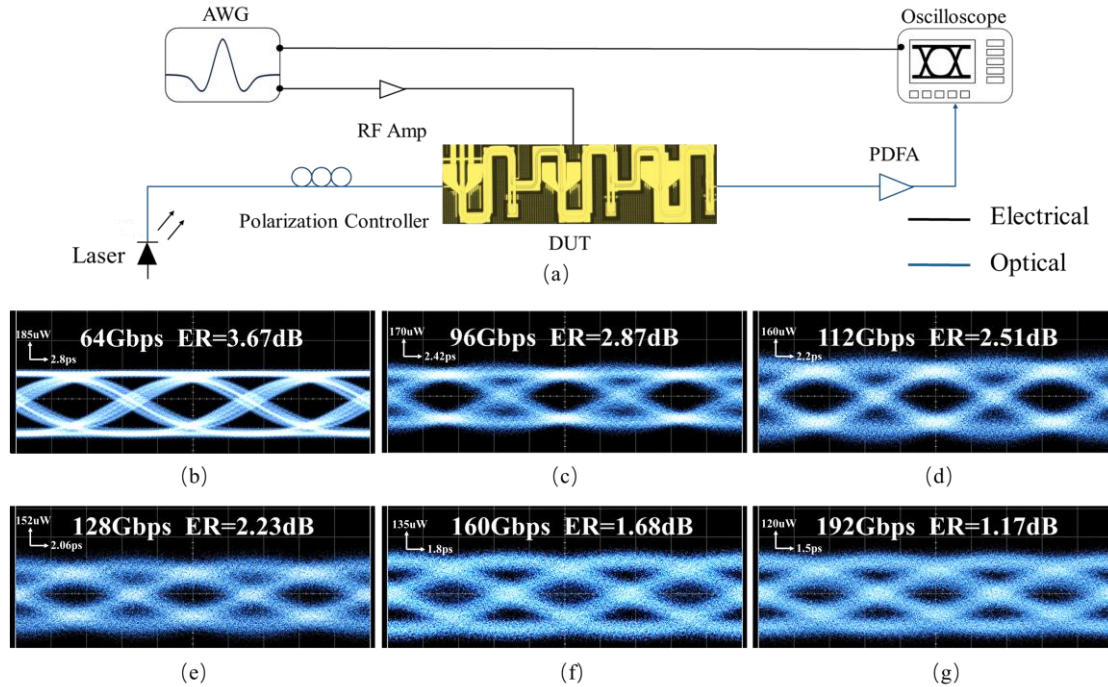


Fig. 3. (a) Experimental setup for the MZM eye diagram measurement. (b)-(g) Measured optical eye diagrams at 64, 96, 112, 128, 160, and 192 Gbaud OOK signals.

Fig. 3(a) illustrates the schematic of the large signal experimental setup for the serpentine segmented MZM. RF signals are generated by channels of a 256 GSa/s arbitrary waveform generator (AWG), amplified by a 67 GHz, 11 dB amplifier, and applied to a single segment of the MZM via a 67 GHz GS RF probe. The output optical signal from the MZM is amplified through a praseodymium-doped fiber amplifier (PDFA) and detected with a 110 GHz oscilloscope. Fig. 3 (b)-(g) show the measured optical eye diagrams of the MZM for 64, 96, 112, 128, 160, and 192 Gbaud with PRBS7 OOK signals. The corresponding extinction ratios are 3.67 dB, 2.87 dB, 2.51 dB, 2.23 dB, 1.68 dB and 1.17 dB, respectively.

4. Conclusion

We report a serpentine segmented MZM with a footprint of $2.25 \text{ mm} \times 0.45 \text{ mm}$, achieving a 1dB EO bandwidth exceeding 67 GHz under 0 V bias. Experimental results demonstrate 192 Gbaud NRZ optical transmission. By employing a serpentine segmented design, the modulator not only achieves higher bandwidth but also significantly enhances compactness and integration density. This structural optimization meets the demands of optical communication systems for high integration and data rates, highlighting the promising potential of silicon photonics for 400 Gbps/ λ data transmission.

5. Acknowledgement

This work was supported by the National Key Research and Development Program of China (2022YFB2803100).

6. References

- [1] Y. Shi, Y. Zhang, Y. Wan, et al., "Silicon photonics for high-capacity data communications," *Photonics Research*, 10, A106–A134 (2022).
- [2] G. T. Reed, G. Mashanovich, F. Y. Gardes, et al., "Silicon optical modulators," *Nat. Photonics* 4, 518–526 (2010).
- [3] Q. Cheng, M. Bahadori, M. Glick, et al., "Recent advances in optical technologies for data centers: a review," *Optica* 5, 1354–1370 (2018).
- [4] R. Soref, B. Bennett, "Electrooptical effects in silicon," *IEEE J. Quantum Electron.* 23(1), pp. 123–129, (1987).
- [5] C. Han, Z. Zheng, H. Shu, et al., "Slow-light silicon modulator with 110-GHz bandwidth," *Sci. Adv.* 9, eadi5339 (2023).
- [6] A. Mohammadi, Z. Zheng, X. Zhang, et al., "Segmented silicon modulator with a bandwidth beyond 67 GHz for high-speed signaling," *Journal of Lightwave Technology*, PP, 1–8 (2023).
- [7] H. Yue, J. Fu, H. Zhang, et al., "Silicon modulator exceeding 110 GHz using tunable time-frequency equalization," *Optica* 12, 203–215 (2025).
- [8] A. Mohammadi, S. Levasseur, W. Shi, "Impedance Peaking in Silicon Traveling Wave Mach-Zehnder Modulators," 2023 Conference on Lasers and Electro-Optics (CLEO), 2023, pp. 1–2.
- [9] A. Mohammadi, L. A. Rusch, W. Shi, "Bandwidth enhancement of silicon traveling-wave modulators using distributed micro-capacitors," in 49th European Conference on Optical Communications (ECOC 2023), 2023, pp. 1512–1515.